

Fairness and Bias in Artificial Intelligence

A Brief Survey of Sources, Impacts, and Mitigation Strategies

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Paper at a Glance

This survey provides a comprehensive overview of **bias and fairness in AI systems**, covering three core pillars: **(1) Sources of bias** (data, algorithmic, and user-level), **(2) Societal impacts** (discrimination, inequality, ethical concerns), and **(3) Mitigation strategies** (pre-processing, model selection, post-processing). Special attention is given to **generative AI (GenAI) bias**, an emergent area where models reproduce and amplify societal stereotypes.

Sources of Bias in AI

Definition

Bias is defined as a *systematic error in decision-making processes that results in unfair outcomes*. In AI, bias can stem from data collection, algorithm design, or human interpretation. Machine learning models learn and replicate patterns of bias present in their training data.

Types of Bias

Type	Description	Example
Sampling Bias	Training data not representative of the target population	Facial recognition trained mostly on white individuals
Algorithmic Bias	Algorithm design prioritizes certain attributes unfairly	Algorithm penalizing age/gender in hiring decisions
Representation Bias	Dataset doesn't accurately represent the population	Medical dataset underrepresenting women
Confirmation Bias	System confirms pre-existing biases of creators/users	Job-success prediction based on manager's biases
Measurement Bias	Data collection systematically over/under-represents groups	Survey over-sampling urban vs. rural residents
Interaction Bias	AI interacts with humans in a biased manner	Chatbot responding differently to men and women
Generative Bias	GenAI outputs disproportionately reflect skewed patterns	Text-to-image models depicting CEOs predominantly as men

Real-World Examples

COMPAS Recidivism System

A ProPublica study found the COMPAS criminal risk-assessment tool was biased against African-American defendants, labeling them high-risk at greater rates than white defendants with similar profiles (Angwin et al., 2016).

Healthcare Mortality Prediction

An AI system for predicting patient mortality assigned higher-risk scores to African-American patients even when controlling for age and health status, risking unequal access to care (Obermeyer et al., 2019).

Facial Recognition & Generative AI

NIST found facial recognition accuracy was significantly lower for darker skin tones (Schwartz et al., 2022). Text-to-image models (Stable Diffusion, DALL-E, Midjourney) depicted CEOs mostly as men and criminals disproportionately as people of color (Nicoletti & Bass, 2023).

Impacts of Bias in AI

Negative Societal Impacts

Biased AI systems produce harms at both the individual and societal level:

- **Discrimination:** Perpetuates and amplifies existing inequalities — e.g., biased criminal justice algorithms, unfair credit scoring (Sweeney, 2013; Dwork et al., 2012).
- **Restricted access to services:** People of color and lower-income individuals face barriers in healthcare, finance, and employment.
- **Gender stereotyping:** Facial recognition systems trained predominantly on men fail to recognize female faces accurately (Buolamwini & Gebru, 2018).
- **New discrimination forms:** Bias based on ethnicity or physical appearance reinforced through GenAI content generation.
- **Cultural entrenchment:** Widespread biased AI can normalize discriminatory narratives and hinder progress toward equality (Ferrara, 2023b).

Ethical Implications

Concern	Details
Discrimination Risk	Race, gender, age, disability — biased systems reinforce marginalization
Developer Responsibility	Developers, companies, and governments share accountability for discriminatory outcomes (Mittelstadt et al., 2016)
Erosion of Trust	Biased AI undermines public confidence in technology, hampering adoption
Loss of Autonomy	Biased hiring/lending algorithms limit individual freedom and opportunity

Core Ethical Principle

Addressing AI bias requires ethical guidelines and regulatory frameworks that enforce **fairness, transparency, and accountability** throughout the AI development lifecycle.

Mitigation Strategies for Bias

Overview of Approaches

Approach	Description	Techniques	Limitations
Pre-processing Data	Address biases before model training	Oversampling, undersampling, data augmentation, adversarial debiasing	Time-consuming; may be ineffective if source data is already biased
Model Selection	Choose/design models that prioritize fairness	Regularization, ensemble methods, group/individual fairness criteria	Lack of consensus on what constitutes fairness
Post-processing	Adjust model outputs after inference	Equalized odds calibration (Hardt et al., 2016)	Complex; requires large additional datasets

Challenges

- **Data scarcity:** Diverse and representative training data is difficult to collect, especially for rare or sensitive events.
- **Detection difficulty:** Algorithmic bias is hard to isolate when arising from multiple interacting sources.
- **Fairness–accuracy trade-offs:** Enforcing equal treatment for all groups may reduce overall model accuracy.
- **Ethical prioritization:** It is unclear whether all biases should be treated equally or whether historically marginalized groups deserve priority.

Generative AI — Special Considerations

For GenAI, a *holistic strategy* is required:

1. **Pre-processing:** Curate diverse, representative datasets from varied cultural and demographic sources.
2. **Model design:** Use adversarial training to test for bias in generated outputs; prefer transparent algorithms.
3. **Post-processing:** Apply additional filters or transfer learning to correct biased outputs.
4. **Continuous monitoring:** Regular audits and feedback loops to maintain fairness over time.

Fairness in AI

Definition and Key Distinction

Bias	Fairness
Systematic deviation from true/expected outcome	Absence of discrimination or favoritism
Can be <i>unintentional</i>	Inherently <i>deliberate and intentional</i>
Can be positive or negative	Concerned only with <i>negative</i> bias
Primarily a <i>technical</i> issue	Primarily a <i>social and ethical</i> issue

Types of Fairness

Type	Description	Example / Reference
Group Fairness	Equal/proportional treatment across demographic groups; includes demographic parity, disparate mistreatment, equal opportunity	Demographic parity in classifiers (Kamiran & Calders, 2012)
Individual Fairness	Similar individuals treated similarly regardless of group membership	Similarity-based measures (Dwork et al., 2012)
Counterfactual Fairness	Same decision made regardless of group attributes in hypothetical scenarios	Kusner et al. (2017)
Procedural Fairness	Decision-making process itself is transparent and fair	Transparent AI governance processes
Causal Fairness	System does not perpetuate historical biases	Kleinberg et al. (2018)

Mitigation Strategies for Fairness

Approach	Description	Key Challenge
Group Fairness	Re-sampling, pre/post-processing to prevent group-level discrimination	May cause intra-group inequality; ignores intersectionality
Individual Fairness	Counterfactual/causal techniques ensuring equal treatment per individual	May not address systemic, group-level biases
Transparency	Making decision logic visible to users	Defining universally agreed fairness criteria
Accountability	Holding developers responsible for harmful outcomes	Assigning responsibility in complex systems
Explainability	Providing understandable justifications for AI decisions	Complexity of human behavior modeling
Intersectionality	Accounting for overlapping identity dimensions (race, gender, SES)	Multiple interacting fairness dimensions

Conclusions & Future Directions

Key Conclusions

1. AI systems, if not carefully designed and audited, **perpetuate and amplify** existing societal biases — especially related to race and gender.
2. **Generative AI** poses emergent bias risks that require dedicated holistic mitigation pipelines.
3. No single mitigation strategy is sufficient; a **multi-layered, interdisciplinary approach** (pre-processing + model selection + post-processing + monitoring) is essential.
4. **Diverse teams** in AI development are crucial for identifying hidden biases.
5. **Ethical and legal frameworks** must treat privacy, transparency, and accountability as foundational — not afterthoughts.

Future Research Priorities

- Diversification of training data, especially for underrepresented demographics.
- Developing bias-detection frameworks tailored to generative models.
- Transparent documentation standards for datasets and model choices (e.g., Datasheets for Datasets — Gebru et al., 2021).
- Exploring implications of synthetic-reality generation on societal norms.
- Robust legal and ethical governance structures for the full AI development lifecycle

(Wachter et al., 2018).

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